HERIOT-WATT UNIVERSITY

SCHOOL OF MATHEMATICAL AND COMPUTER SCIENCES

DEPARTMENT OF ACTUARIAL MATHEMATICS AND STATISTICS

F72ZA1
FINANCIAL MATHEMATICS 1

7 December 2007

Time allowed 2 hours

Attempt all **7** questions

Total Marks 60

Only approved calculators may be used.

NUMERICAL SOLUTIONS

1. Fill in the missing values in the following table:

i	v	d	$i^{(12)}$	$d^{(4)}$
0.05	0.952381	0.047619	0.048889	0.048494

2. Solving $75(1.06)^t = 100$, the fund reaches £75,000 at time $20-t$, where $t = 4.937144$ years.

   Hence $X(1.05)^{10} + X = 75,000(1.05)^{4.937144-10} \Rightarrow X = £22,284.85$.

3. (a) Bookwork proofs

   (b) (i) 8.5595
   (ii) 5.0458
   (iii) 56.4454

   (c) £7,107.00

4. (a) (i) £8,581.05
   (ii) £81,062.16
   (iii) £5,674.35

   (b) Need to find the smallest integer n such that $7,500a_{\overline{n}|}^{0.07} \geq 78,155.46 \Rightarrow n = 20$ years from now, an extension of 5 year to the original term.

5. (a) Bookwork

   (b) Yields on A, B and C are 0.1 *per annum*, 0.11 *per annum* and (by trial and error and linear interpolation) approximately 0.124 *per annum*, respectively. Hence we prefer C.

6. (a) £125.72 (note CGT is not payable)

   (b) The redemption yield is 4% *per annum*, from (a). If A accepts B's offer the yield on the completed transaction will be:

$$\frac{4 \times 0.8}{125.72} = 0.025453 \text{ } \textit{per half-year} = 0.0516 \text{ } \textit{per annum}.$$

   Hence A should accept B's offer.

7. (a) The yield in 2007–8 is $i_1 = -0.1885$. The yield in 2009–10 is $i_2 = 0160397$. Hence LIRR is given as the solution of:

$$(1 + \text{LIRR})^{3.5} = (1 + i_1)^2(1 + i_2)^{1.5} \Rightarrow \text{LIRR} = -0.054.$$

   (b) TWRR is given by:

$$\left(\frac{2.25}{2.15}.\frac{1.7}{1.9}.\frac{2.0}{1.6}\right)^{1/3.5} - 1 = 0.046.$$